HARSHITA PURANIK

ROBOTICS & AI | COMPUTER VISION | EMBEDDED SYSTEMS | AUTONOMOUS ROBOTS | ROS2
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SUMMARY

Experienced in embedded systems, robot manipulation, and computer vision. Skilled in developing robotic solutions through perception, control, and hardware-software integration. Eager to explore applications of robotic systems in various domains.

SKILLS

- **Programming** : Python, C/C++, SQL
- **Microcontrollers** : Arduino, ESP32, STM32
- **Control** : PID, Q-learning (RL), Kinematics, Odometry
- **Tools** : Gazebo, CoppeliaSim, SolidWorks
- **Communication** : UART, I2C, SPI, BLE, WiFi, CAN
- **IoT** : System Integration, Sensor Fusion
- **CV** : OpenCV, Mediapipe, YOLOv11, SAM2
- **Framework** : ROS2 Humble

WORK EXPERIENCE

Embedded System Intern | SOUL3 | July 2026 - Present

- Working on exoskeleton prototype.

Computer Vision Intern | GrayMatter Robotics | May 2026 - July 2026

- Developed computer vision model using YOLOv11 for detection and SAM2 pipeline for segmentation and accuracy to detect liquids used in modern kitchen, paired with physical sensors for classifying visually similar liquids. Simulation in Gazebo using ROS2 nodes

Autonomous Systems Dept. | Team ETA | Mar 2026 - Present

- Working on simulation test environments using Gazebo.

Embedded System Dept. | Team KJSSE Robocon | Oct 2024 - Jan 2026

- Worked as a member of embedded coding department, dealing with microcontrollers, sensors, actuators, communication protocols and control systems on daily basis.
- Participated in DD Robocon 2025, organised by IIT Delhi.

EDUCATION

B.Tech Robotics & AI **2024 - 2028**
K. J. Somaiya School of Engineering, Vidyavihar
Average CGPA : 8.43

PROJECTS & COMPETITIONS

CropDrop Robot | E-yantra Robotics Competition 2025

- Built a STM32 based autonomous line follower combining sensor integration, PID, Reinforcement Learning (Q-learning algorithm) and CoppelisSim for testing environment.

TerraBot | Agri-Tech Hackathon 2026

- Our team developed an agricultural robot for plant disease detection using CNN model and real-time field monitoring using sensor integration.
- Wireless communication between ESP32 and software UI for easy operation of the robot.

Hand Gesture Controlled Car

- Developed a hand gesture recognition model using openCV and mediapipe and mapped the detection outputs with motor control commands through WiFi communication between Python and Arduino IDE.

Swarm based Marine Robotics System | MechTech Ideation 2026

- Designed a bio-robotics inspired fish-like structure according to each level of depth in the water bodies for surveillance and waste detection. Suggested acoustic communication between swarms and a combination to 2 algorithms - Boids/Flocking Algorithm and Partical Swarm Optimization to ensure navigation and formation of the swarms.

Smart Safari System | Unplugged 3.0

- Developed animal detection model using YOLO, software UI using Python and line follower mechanism using PID and sensor fusion, aiming towards elevating the experience of existing Safari System.